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A Novel Plasma Based Technology for Efficient Produced Water Treatment, Recycle, and Reuse

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Abstract

Advanced oxidative processes (AOPs) are increasingly recognized for their efficacy in degrading persistent organic pollutants, pathogens, and other contaminants in water treatment. Among AOP techniques, plasma chemistry stands out due to its synergistic integration of physical (e.g., UV photons, shockwaves, electric fields) and chemical (e.g., reactive oxygen and nitrogen species, ozone, hydroxyl radicals) effects, enabling efficient degradation of complex pollutants. This study focuses on applying plasma-based AOPs for the treatment of produced water (PW) from oil and gas operations, which contains contaminants including hydrocarbons, heavy metals, and dissolved solids that challenge conventional methods. We aim to design, optimize, and evaluate a scalable and energy-efficient plasma-driven AOP system for PW remediation, emphasizing regulatory compliance for safe discharge or reuse. By leveraging plasma's multi-reactive mechanisms, this approach seeks to enhance contaminant removal efficiency while minimizing secondary waste generation, offering a sustainable solution for PW recycling in upstream applications such as hydraulic fracturing, enhanced oil recovery, or irrigation. In this work the underwater diaphragm discharge plasma water treatment device was designed, and the ideal electric discharge parameters were found. The research revealed that plasma treatment produces iron oxide nanoparticles ranging from 100 ± 20 to 657 ± 120 in size. The investigations on reservoir water solutions showed that iron oxide nanoparticles absorb dissolved salts on their surfaces and plasma activation decreases interfacial tension (on the hexane border). By changing the flow-type cell architecture, iron oxide nanostructure synthesis may boost energy efficiency to 18.75 g/kWh. The treatment by this discharge allow to remove oil contaminations and reduce the salinity.

Introduction

Conventional water purification methods such as sorption, photocatalytic degradation, flotation, membrane separation, and chemical precipitation often involve multistage processes and excessive chemical reagent use, leading to secondary pollution and increased costs (Manakhov 2022). To address these limitations, advanced oxidation technologies (AOTs) have emerged as promising alternatives, leveraging highly reactive species to degrade pollutants without extensive chemical inputs. Among these, low-temperature plasma in

contact with liquids has gained attention due to its ability to generate a synergistic mix of oxidative and reductive species, enabling efficient contaminant breakdown (Thagard 2018).

Electrical discharges in aqueous environments produce nonthermal plasma with distinct characteristics depending on their configuration. For instance, glow discharges generate gas-phase plasma rich in reactive species, whereas underwater discharges enhance liquid-plasma interactions through convective flows and cavitation effects (Tang 2009, Ghodbane 2014). These processes facilitate the formation of reactive oxygen and nitrogen species (e.g., hydroxyl radicals, hydrogen peroxide, ozone) as well as reductive agents (e.g., hydrated electrons, hydrogen atoms), driving redox reactions that degrade dissolved contaminants (Sirotkin 2020). The oxidative potential of hydroxyl radicals (-2.7 V) and hydrogen peroxide plays a crucial role in breaking down organic pollutants, while reductive species contribute to the transformation of inorganic compounds (Zeghioud 2020).

Extensive research has demonstrated the effectiveness of plasma-based treatment in degrading single-dye solutions, with methylene blue (a thiazine dye) and azo dyes being frequently studied due to their prevalence in industrial effluents (Mohammadi 2016). Additionally, anthraquinone dyes, known for their carcinogenic properties, have been targeted for decomposition via plasma-induced reactions. However, real industrial wastewater typically contains complex mixtures of dyes and co-pollutants, presenting a more challenging remediation scenario. Recent studies indicate that the degradation kinetics of dye mixtures may differ significantly from single-dye systems, with some cases showing accelerated removal rates in multicomponent solutions (Khlyustova 2019). These findings suggest potential synergistic effects, where the presence of multiple dyes influences reaction pathways and degradation efficiency.

Despite progress in understanding plasma-driven dye degradation, gaps remain in optimizing treatment systems for complex, real-world wastewater. This study explores the application of low-temperature plasma for the treatment of mixed-dye effluents, examining the interplay between different dye classes and the role of reactive species in simultaneous degradation. By evaluating key parameters such as discharge type, solution composition, and reaction kinetics, this work aims to enhance the scalability and efficiency of plasma-based AOTs for industrial wastewater treatment.

Indeed, plasma-solution systems have emerged as a promising approach for the synthesis of nanostructures, offering advantages such as reductant-free processing, simple experimental design, and continuous synthesis under plasma exposure. These systems enable the direct generation of reducing agents during nanoparticle formation, distinguishing them from conventional liquid-phase methods. Recent studies highlight the application of plasma-liquid interactions for the synthesis of metal oxides (e.g., Al_2O_3 , CuO , TiO_2 , ZnO), utilizing either metal salt solutions or electrodes as metal sources. By adjusting plasma parameters and experimental conditions, the phase composition of synthesized powders can be controlled. For instance, variations in discharge media (e.g., H_2O_2 solutions, oxygen bubbling) and plasma settings influence the crystallinity and phase distribution of TiO_2 , with some systems yielding defective or mixed-phase structures. Atmospheric pressure glow discharge with a liquid cathode has also been employed to produce transition metal oxides, though impurities from precursor salts or hydroxides may persist. Plasma-chemical synthesis in aqueous media ensures rapid quenching due to high thermal gradients, facilitating the formation of pure oxides without precursor residues.

Nevertheless, the plasma treatment processes were never employed for the produced water treatment. In this work, for the first time the plasma treatment of the synthetic wastewater created can effectively break down organic pollutants and remove petroleum contaminants. The developed upscaled plasma reactor is a modular, flow-type system with a capacity of up to 9.5 L and a length adjustable to varying solution volumes. The reactor enables simultaneous underwater plasma discharges via up to 8 electrodes powered by a high-voltage circuit. Testing revealed its effectiveness in altering IFT between oil and displacing fluid, with notable reductions under optimized conditions. Environmental uses included quickly breaking down organic dyes (like crystal violet and Rhodamine B) through plasma-induced breakdown and changing motor

oil mixtures into forms that can be filtered out. The technology highlights scalability and adaptability for the treatment of produced and polluted water, with performance influenced by treatment duration, diaphragm configuration, and solution properties.

Description and Application of Equipment and Processes

Equipment

The testing of the modeled produced water treatment was performed using in-house made flow-type prototype that consists of three parts: a reservoir with the initial high-salinity water, a plasma reactor, and a reservoir with treated water (Fig. 1). The plasma reactor and current leads are made of quartz glass to visualize the combustion process of an underwater discharge. The tanks are made of polypropylene pipes. The total volume of the solution being processed was 7.5 L, with a total capacity of 10 L. The solution flow was created by a ZUFER peristaltic pump with a pumping speed of 500 ml/min (the maximum pumping speed is 640 ml/min). The electrical circuit of an underwater diaphragm discharge is shown in Fig. 2. Voltage is supplied to the electrodes from a high-voltage transformer (UM-1.25/10-UHL1) through an autotransformer. The average power consumption is recorded by a PC-7 wattmeter (100-130 W). The design of the prototype installation has two current inputs, which makes it possible to ignite a discharge using two diaphragms.



Figure 1—Prototype of flow-type installation.

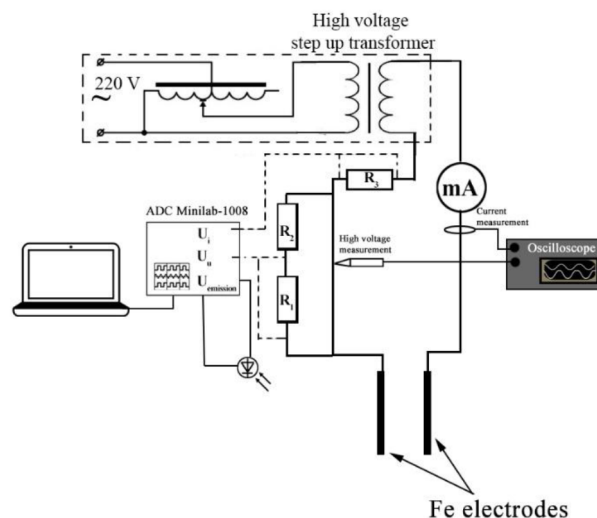


Figure 2—Electrical scheme of an underwater discharge.

Procedure

A plasmachemical cell is a structure of two glass containers in the form of cylinders, inserted one into the other, and providing for the circulation of the solution using a peristaltic pump. Solutions of two compositions were prepared for the following testing. The concentrations of the main salts dissolved are presented in Table 1.

Table 1—The concentrations of dissolved salts in tested water samples.

Solution I		Solution II	
Compound	Concentration, g/L	Compound	Concentration, g/L
NaCl	0.5-21.5	NaCl	0.5-21
CaCl ₂	0.3-0.6	CaCl ₂	0.3-5.748
MgCl ₂	0.2	MgCl ₂	0.2-0.947
Na ₂ SO ₄	0.5-1	K ₂ SO ₄	0.5-0.975
		NaHCO ₃	0.27

To prepare solutions of a given concentration, the salts of NaCl, CaCl₂, MgCl₂, Na₂SO₄/K₂SO₄, NaHCO₃ (ChemReaktiv, Russia) were dissolved in distilled.

A solution with a specified salinity was subjected to an underwater diaphragm discharge from an alternating current source operating at a frequency of 50 Hz. Electrodes composed of iron wires with a diameter of 0.6 mm were utilized. The duration of the therapy was 60 minutes. The discharge's electrical properties were documented utilizing an oscilloscope (ATAKOM ADS 2072, Russia) and an analog-to-digital converter (MiniLab 1008, USA). A comprehensive calculating approach for discharge parameters, including efficiency, is presented below.

The discharge current and voltage drop results were documented using the ADS-2072 digital oscilloscope and subsequently analyzed on a computer. A voltage divider with a 1:1000 division ratio was employed to measure the voltage drop. The data from the voltage divider, ballast, and photodiode were digitized using the ADC "Measurement MiniLab 1008" and sent to a personal computer. The digitized data is captured on a personal computer using specialized software known as "TracerDAQ". The conversion of the received signals can be taken into account by measuring the values of the resistances of the resistors R_1 , R_2 , R_3 . Thus, the value of potential between the electrodes can be calculated as:

$$U = U_u \frac{R_1 + R_2}{R_2}, \quad \text{Equation (1)}$$

The value of electrical current in the circuit is:

$$i = \frac{U_i}{R_3} \quad \text{Equation (2)}$$

The value of discharge energy is:

$$E = \int_{t_1}^{t_2} U(t)i(t)dt, \quad \text{Equation (3)}$$

where U_u is the voltage drop on the resistor R_2 , U_i is the voltage drop on the ballast resistor R_3 , R_1 is the resistor of 42.0 MOhm, R_2 is the resistor of 42.0 kOhm, R_3 is the resistor of 3.8 Ohm, t_1 , t_2 are the initial and final moments of discharge.

The average values of the current, voltage of the discharge, and power dissipation were obtained by averaging measurements for 30 seconds. Thus, the average voltage is:

$$\left\langle U \right\rangle = \frac{1}{T} \int_t^{T+t} U_i(t)dt \quad \text{Equation (4)}$$

The average current is:

$$\left\langle i \right\rangle = \frac{1}{T} \int_t^{T+t} i_i(t)dt \quad \text{Equation (5)}$$

The average power is:

$$\left\langle P \right\rangle = \frac{1}{T} \int_t^{T+t} U_i(t)i_i(t)dt, \quad \text{Equation (6)}$$

where t is the initial integration time, T is the integration period, U_i is the instantaneous value of the voltage on the electrodes, i_i is the instantaneous value of the current in the circuit.

The total energy dissipated in the discharge cell is the sum of the energy of the electrical current (without the formation of discharges E_0) and the energy dissipated in the discharges E_p :

$$E_{total} = E_0 + E_p \quad \text{Equation (7)}$$

The energy of a single discharge can be calculated by the formula:

$$E = P \times t, \quad \text{Equation (8)}$$

where P is the discharge power; t is the discharge duration.

Therefore, the total energy of the discharges:

$$E_p = \sum_{i=0}^n P_d \times t_{id}, \quad \text{Equation (9)}$$

where P_d is the power of a single discharge, t_{id} is the duration of a single discharge.

The discharge power dissipated in the liquid is the sum of the power dissipated by the current flow in the solution and the power dissipated in the discharge. Therefore, each amplitude value of the discharge power should be subtracted from the corresponding value of the amplitude value of the power dissipated in the cell without the formation of a discharge.

To simplify the calculations, we use the average amplitude value of the power dissipated during the discharge and subtract the average amplitude value of the power dissipated in the cell without generating a discharge. Then the final expression will have the form:

$$E_d = \frac{1}{2} \left(\sum_{i=0}^n P_{adi} - P_p \right) \times \sum_{i=0}^n (t_1 - t_0), \quad \text{Equation (10)}$$

where P_{adi} is the total value of the power dissipated by the flow of current in the solution and the power dissipated in the discharge, P_p is the power dissipated in the cell without the formation of discharges, $t_1 - t_0$ is the duration of the discharge.

Data and Results

Adjustment of the electrical characteristics for produced water treatment

The average values of the discharge parameters are presented in Table 2, and the current and voltage oscillograms are shown in Fig. 1. The data obtained showed that the composition of the solution (the presence of potassium ions) does not have a significant effect on the average values of electrical parameters. When the concentration varies towards a decrease in total salinity, the average values of the power and frequency of discharge occurrence increase (Fig.3.).

Table 2—Electrical characteristics of underwater AC diaphragm discharge.

Parameter	Solution I (C=25 g/L)	Solution II (C=27 g/L)
Average single discharge power, W	44	30.85
Discharge frequency, Hz	2	1.2
Efficiency, %	65.45	59.88

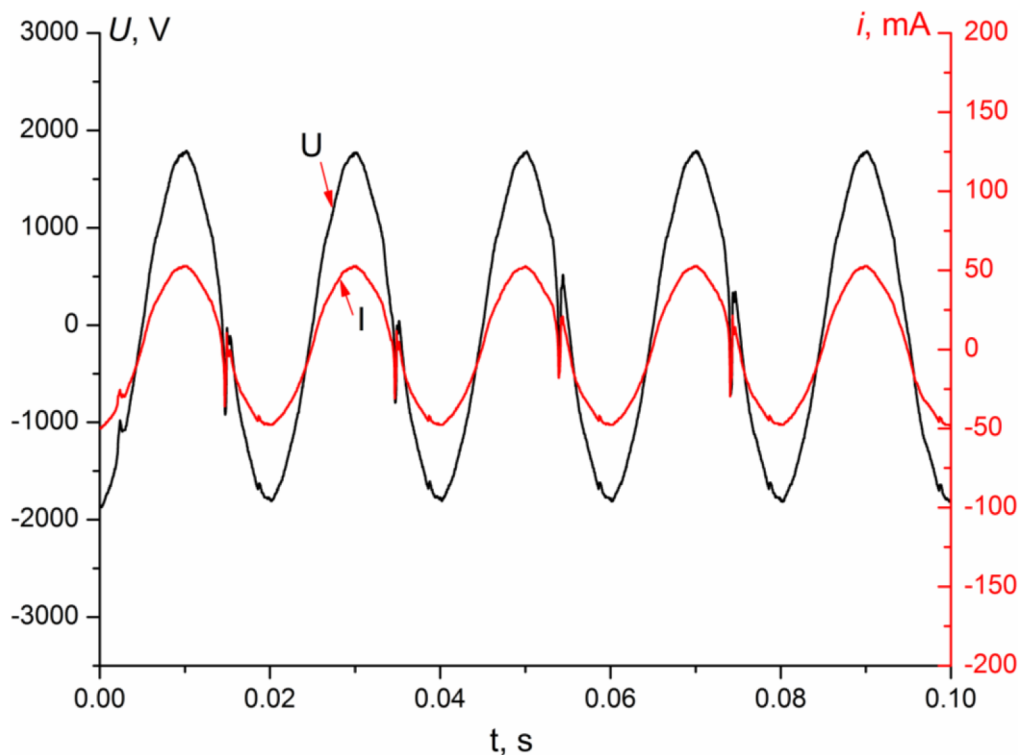


Figure 3—Current and voltage waveforms of underwater AC diaphragm discharge.

Before and after plasma treatment, the electrical conductivity and interfacial tension between working solutions and hexane were measured. Electrical conductivity was recorded using an Anion 4100 (Russia) conductometer. Interfacial tension measurements were carried out using the ring pull-off method on a K6 tensiometer (Kruss GmbH, Germany). Hexane (LenReaktiv, Russia) was used as a model of oil. The measurement results are shown in Table 3.

Table 3—Changes in electrical conductivity and interfacial tension as a result of plasma treatment.

Solution	C, g/L	Conductivity, mS/cm		Interfacial tension, mN/m	
		before	after	before	after
I	25	30.8	29.3	39.0±0	37.0±0
II	27	40.7	44.8	37.63±0.48	41.2±0.91
	25	38.7	47.1	31.0±0	19.67±0.58
	20	34.4	39.2	40.33±0.29	32.75±0.29
	15	28.4	25.76	42.0±0	26.8±1.3

According to the results obtained, underwater plasma treatment leads to an increase in the electrical conductivity of water solutions. The exception is a solution with a low total salinity. Interfacial tension measurement data showed that at a salinity of less than 27 g/l, the underwater plasma treatment leads to a decrease in σ values. Changes in these parameters after plasma treatment are a consequence of the appearance of iron oxides (dispersed phase) in the liquid phase. This hypothesis is confirmed by a change in the color of the solution, which is recorded from the first minutes of the discharge (a visual sign of the appearance of ultrafine particles of iron oxide) (Fig. 4).

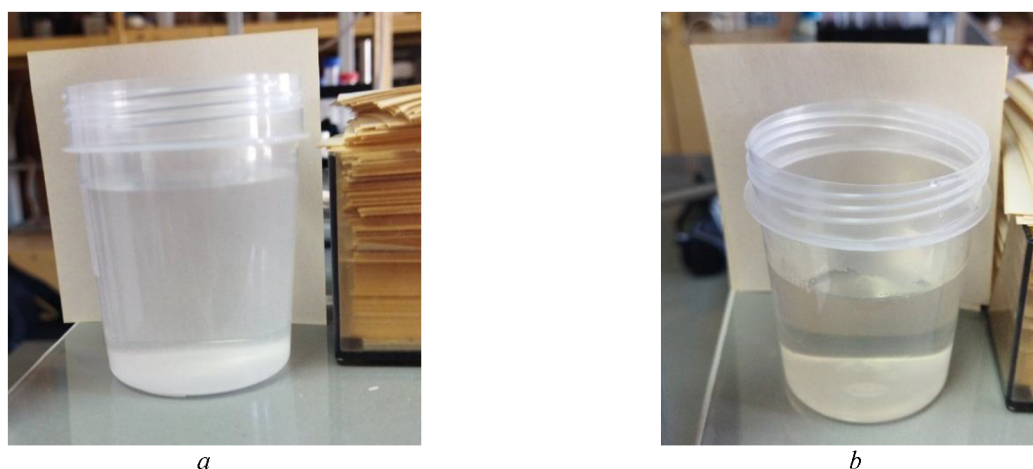


Figure 4—Appearance of the water solution before (a) and after (b) plasma treatment.

Analysis of the precipitated material

The plasma treatment of synthetic produced water led to the formation of significant powder material that was precipitated during plasma treatment.

An analysis of the particle sizes of the dispersed phase was carried out using the dynamic light scattering method (Malvern Zetasizer, UK) (Table 4). There is a tendency for a decrease in particle size with decreasing concentrations of dissolved salts. The effect of the composition of the water solution on particle agglomeration was also recorded.

Table 4—Particles' size of dispersed phase obtained as a result of underwater diaphragm discharge combustion during 60 min.

Solution	C, g/L	Power, W	Average size, nm
I	25	120-125	657±120
II	27	120-125	162.3±50
	25	120-125	150±30
	20	110	117±40
	15	100	100±20

The dispersed phase was separated from the main solution by centrifugation at 6000 rpm for 15 min (UC 6000E, China) and dried at 60°C for 24 hours. Next, structural phase analysis was carried out using diffraction and Raman scattering methods.

The results of X-ray phase analysis showed that the sediment consists of iron oxides (mainly Fe_2O_3 of various polymorphs with magnetic properties) with impurities of iron (II) sulfate and chloride (Fig. 5). Raman vibrational spectroscopy data confirmed that in addition to iron oxides, the sediment contains complex iron-containing salts (Fig. 6). The composition of the tested water solution does not affect the phase composition of the sediment.

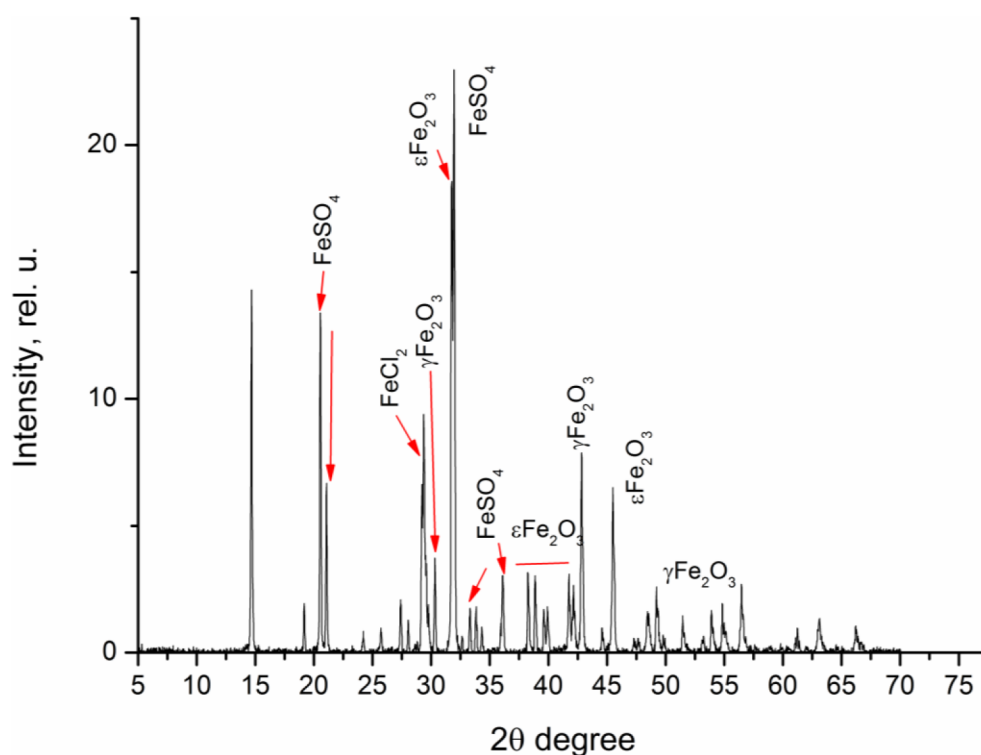


Figure 5—Typical XRD patterns of dispersed phase from the tested water solutions.

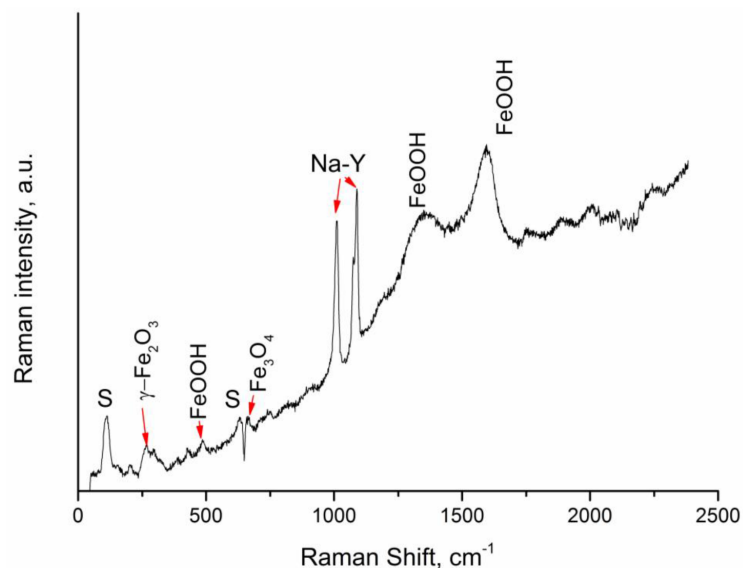


Figure 6—Typical Raman spectra of precipitate from the tested water solutions. It is noted: S – sideronatrite ($\text{Na}_2\text{Fe}(\text{SO}_4)_2(\text{OH}) \cdot 3\text{H}_2\text{O}$), Na-Y – Na-jarosite ($\text{NaFe}_3(\text{SO}_4)_2(\text{OH})_6$).

An image of the surface of the deposit formed during the combustion of a diaphragm discharge and its elemental composition are presented in Fig. 7. The low iron content, according to the analysis results, is explained by the fact that iron oxides, as sorbents, collect dissolved salts on their surface. Nevertheless, this confirms the good ability of iron oxide nanoparticles to clean the produced and waste water from organic contaminants.

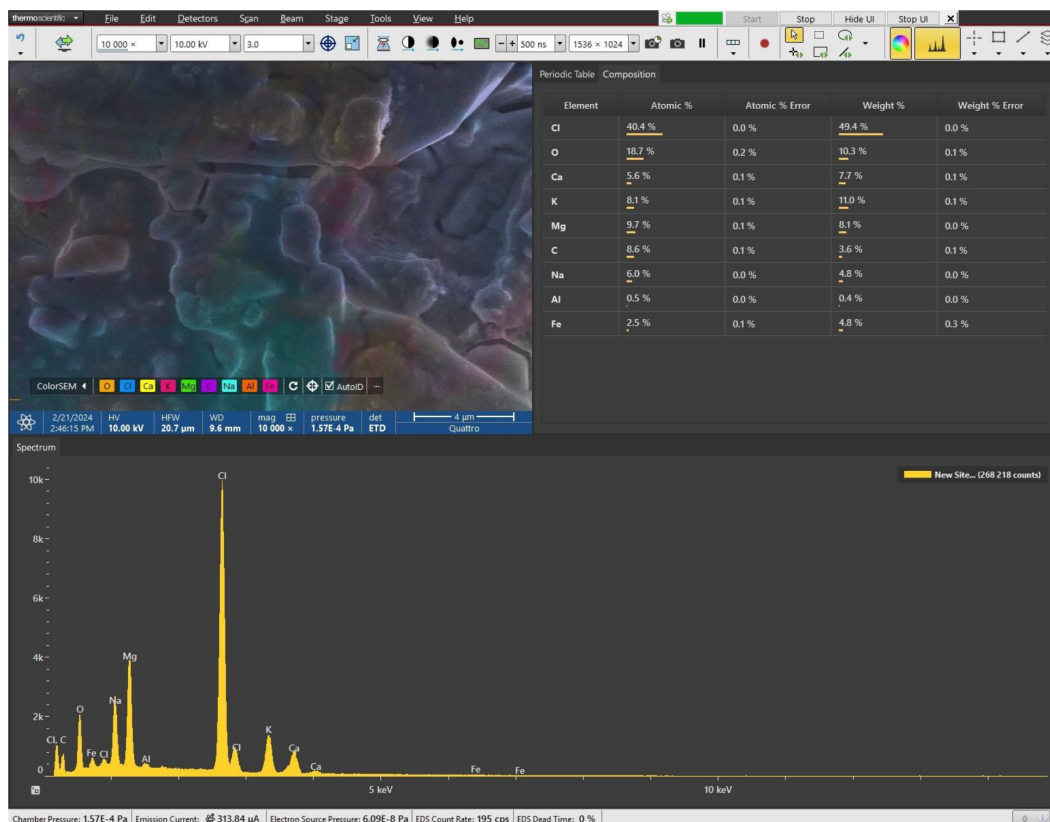


Figure 7—SEM image of precipitate and its elemental analysis data.

During the combustion of the diaphragm discharge, the electrodes are sputtered. This is confirmed by a change in the mass of the electrodes (weighing was carried out using analytical balances AND HR-150AZ, Japan). The change in the mass of the electrodes after the plasma treatment of solutions was 0.0064 g at an expended power of 120 W. We believe that all the sputtered electrode material turns into oxide. Thus, the energy efficiency of producing iron oxides in a flow system of concentrated solutions (models of reservoir water) is 18.75 g/kW h.

Testing of organic contaminations removal.

The effects of underwater discharges are most evident when examining organic dyes (or dye mixtures), where the degradation of large organic molecules is indicated by a visible change in solution color. This color change is quantitatively monitored using spectrophotometry. In the case of petroleum products, plasma treatment can lead to the formation of dispersed fractions with densities that may either exceed or fall below that of the original solution.

Destruction of dye's mixture. A series of experiments on the destruction of different classes of dyes in a solution of their mixture (three dyes) was carried out with varying the number of electrodes included in the electric circuit (the number of underwater discharges). A mixture of three dyes was used as a model solution: crystal violet 4K (a class of active dyes), Rhodamine B (xanthene dyes) and methylene blue (thiazine dye). During the experiment, samples were taken for further spectrophotometric studies. Changes in the intensity at the absorption maxima for each dye were recorded: crystal violet (CV4K) 530 nm, Rhodamine B (RhB) 554 nm and methylene blue (MB) 667 nm. The kinetic curves are shown in Fig. 8.

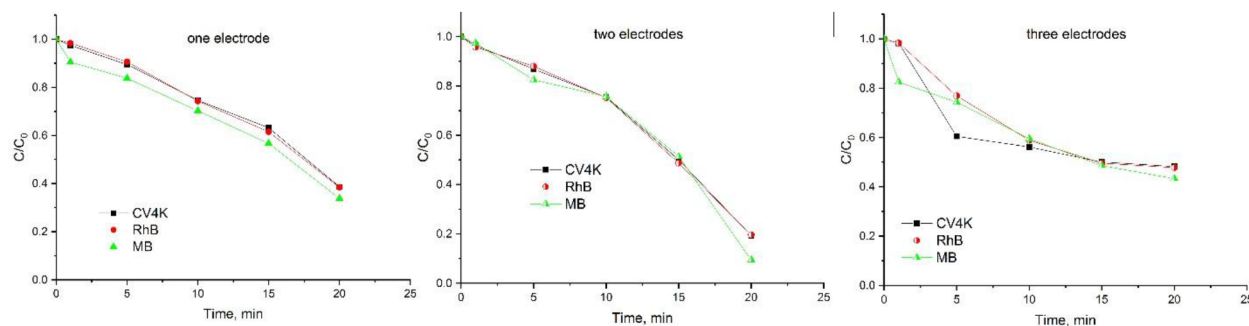


Figure 8—Kinetics of dye destruction in a mixture with varying number of electrodes (discharges). Initial concentration of each dye was 1.2 mg/L, total solution volume was 2 L, and rate of circulation was 600 mL/min

The results indicate that two discharges provide optimal dye removal efficiency while minimizing treatment time. Prolonged plasma exposure leads to accumulation of both saturated and unsaturated carboxylic acids in solution (Khlyustova et al., 2019), suggesting a trade-off between treatment duration and byproduct formation.

Removal of hydrocarbons

The solution of motor oil (synthetic engine oil, 5W-40) emulsion was exposed to the action of underwater discharge. When intensively stirred, the solution is a two-phase system: water and oil bubbles. The action of underwater plasma causes a change in the color of the solution from yellow to brown (Fig. 9a-c), and the appearance of a dispersed phase (Fig. 9d-e). The dispersed phase is easily removed from the surface of the treated solution.

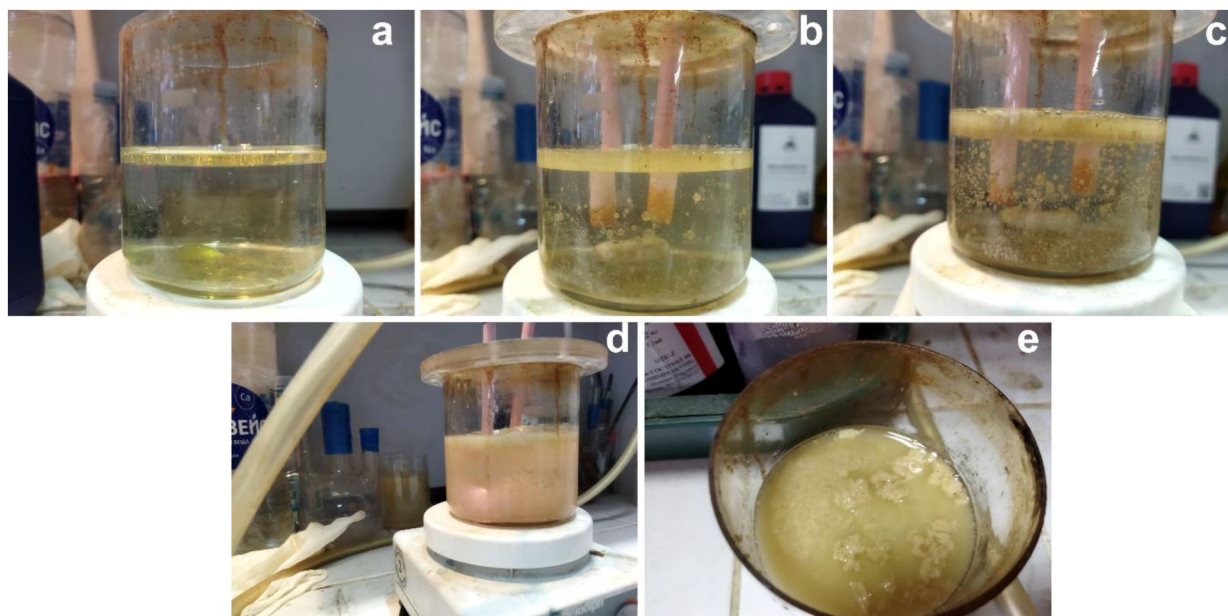


Figure 9—(a) Initial solution of motor oil, (b) solution after 3 min of plasma treatment, (c) solution after 6 min of plasma treatment, (d) appearance view of solution during plasma treatment, and (e) solution surface after plasma treatment (6 min)

Hence, our results demonstrate the effectiveness and versatility of underwater plasma discharges in degrading complex organic pollutants, including dye mixtures and petroleum products. Spectrophotometric analysis of a mixed-dye solution revealed that the optimal degradation efficiency was achieved with two active discharges, beyond which prolonged treatment led to the accumulation of carboxylic acids. In the case of synthetic engine oil, plasma treatment induced visible changes, including a color shift from yellow to brown and the formation of a dispersed phase, which was easily separable from the aqueous solution. These findings highlight the potential of underwater plasma as a versatile method for wastewater treatment, capable of degrading both soluble dyes and emulsified hydrocarbons. Further optimization of discharge parameters could enhance efficiency while minimizing byproduct formation.

Upscaling of plasma treatment process

The applicability of any kind of the water treatment technologies depends on the cost-effective upscaling. Thus we have developed and tested upscaled modular reactor that can be employed in different kinds of industries. The upscaled plasma reactor offers potential for industries managing produced water and wastewater with organic contaminants. By degrading complex dyes (e.g., crystal violet, Rhodamine B) within minutes and converting motor oil emulsions into separable phases, the technology reduces reliance on chemical additives and secondary treatment steps, lowering operational costs. Modular design of the upscaled reactor allows customization for varying volumes, making it adaptable to industrial applications. The technology generates plasma-activated water with a unique composition that enhances oil mobility, improves micropore and volumetric sweep efficiency, and ultimately increases oil displacement. Enhanced oil recovery via interfacial tension reduction (e.g., 20% IFT drop in 20 minutes) after plasma-activated water injection could improve efficiency in petroleum extraction. Furthermore, the electric discharge and plasma generation process inherently heats the injected fluid and reservoir fluids, contributing as an additional thermal recovery mechanism. In summary, the scaled-up plasma treatment technology along with treated produced water and plasma-activated water injection can be effectively applied for sustainable upstream operations. The technology's ability to degrade pollutants (e.g., dyes, oils) positions it as a sustainable alternative in circular economy frameworks, opening markets in environmental remediation, petroleum spill management, and textile wastewater treatment.

The upscaled plasma setup consists of an upscaled flow-type reactor, a circulation pump, and an electrical circuit. Fig. 10 shows a schematic of the upscaled flow-type plasma reactor. The reactor is constructed from polypropylene blocks (fittings) forming a finite-length pipe with an internal diameter of 11 cm. The current leads are ceramic tubes with an outer diameter of 7 mm, an inner diameter of 1.1 mm, and a length of 13 cm each. The electrodes are made of iron wire, measuring 1 mm in diameter and 15 cm in length. The total reactor volume is 9.5 L.

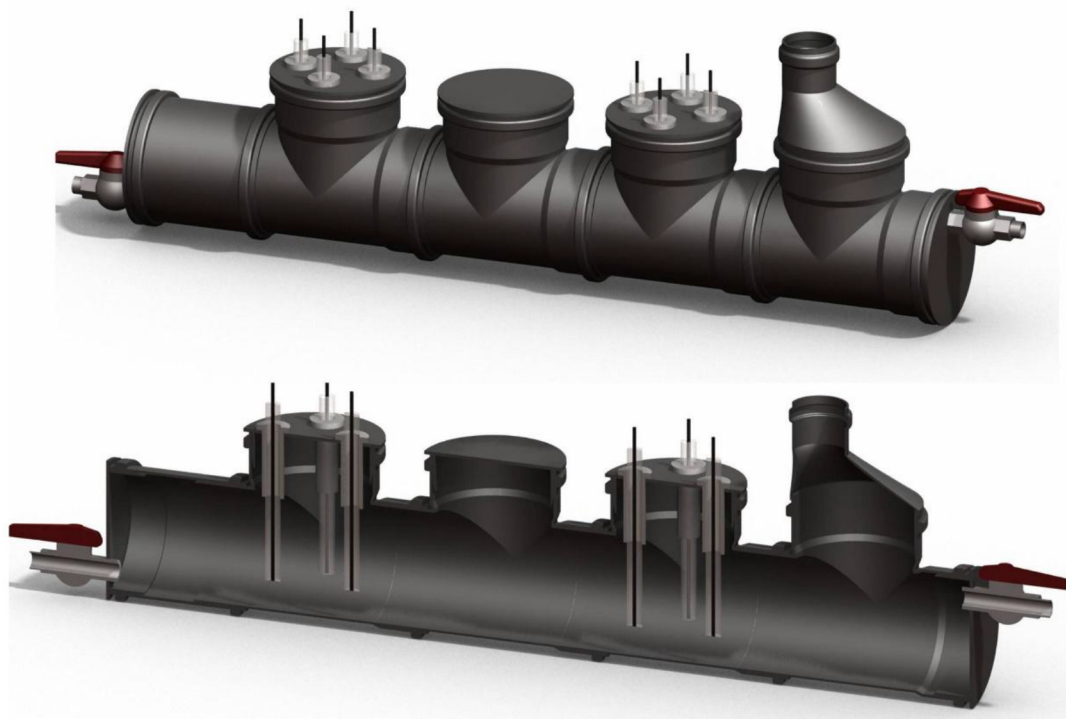


Figure 10—Scheme of the upscaled flow-type plasma reactor

To initiate underwater end discharge using alternating current, a single-phase step-up transformer with a capacity of 1.25 kVA and a transformation ratio of $n=27$ (OM-1.25/10-UKHL1, 50 Hz) was used. The transformer was connected to a household alternating current network (~ 220 V, 50 Hz) through an autotransformer (model TDGC2-3K, ANDELI) for smooth voltage regulation within the circuit. The autotransformer was connected to the network with an insulated wire with copper cores of a cross-section of 2×2.5 mm². To prevent overheating of the autotransformer at peak currents during discharge combustion for the selected transformer power, the autotransformer must have a rated current of at least 9 A.

Measurements of electrical conductivity and interfacial tension on the water solution-hexane boundary were carried out before and after plasma treatment. The experiments with a water solution I showed that at the maximum salinity and treatment time of 60 min, plasma treatment leads to a decrease in the values of interfacial tension in the solution-hexane system (Table 5). In the experiments with water solution II, the reduction effect was noticed when the solution was diluted.

The energy efficiency of producing iron oxides in a prototype installation with two diaphragms and an input power of 120 W is 15.3 g/kW h.

Table 5—Changes in electrical conductivity and interfacial tension during plasma treatment for 60 min.

Solution	C, g/L	Power, W	Conductivity, mS/cm		Interfacial tension, mN/m	
			before	after	before	after
I	25	130-140	28.5	31.2	37.33±0.58	32.0±0
	20	120-125	24.6	27.5	39.0±0	39.0±0
II	27	140-145	30.2	32.7	40±0	40±0
	25	130-140	38.7	47.1	28.75±0.29	33.38±0.48
	20	120-125	25.1	19.8	42±0	23±0

Conclusions

The practical implementation of water treatment technologies hinges on cost-effective scalability. To address this, we developed and tested a modular, upscaled plasma reactor designed for diverse industrial applications. This system demonstrates significant potential for treating produced water and wastewater containing organic contaminants, such as complex dyes (e.g., crystal violet, Rhodamine B) and hydrocarbon emulsions. Key advantages include:

- **Rapid Degradation & Phase Separation:** Organic dyes are degraded within minutes, while motor oil emulsions are converted into separable phases, minimizing the need for chemical additives and secondary treatment.
- **Modular Adaptability:** The reactor's scalable design allows customization for varying treatment volumes, ensuring broad industrial applicability.
- **Improving fluid properties:** Plasma-activated water (PAW) generated by the reactor improves oil mobility through interfacial tension reduction (e.g., 20% decrease within 20 minutes), increased sweep efficiency, and thermal effects from in situ fluid heating.

By integrating pollutant degradation (e.g., dyes, oils) with resource recovery, this technology aligns with circular economy principles, offering sustainable solutions for:

- **Environmental Remediation** (e.g., petroleum spill cleanup),
- **Textile Wastewater Treatment,**
- **Upstream Oilfield Operations** (produced water treatment and PAW-assisted improved recovery).

The dual functionality of the reactor—contaminant destruction and process fluid enhancement—positions it as a versatile tool for industrial water management.

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